

APPENDIX I

THEORETICAL FOUNDATION OF GBSK

In this part, we consider an observed dataset $P = \{p_i\}_{i=1}^n$ of n points independently and identically drawn from a probability distribution $\mathcal{F}$ with density $f(x)$ defined on $\mathbb{R}^d$, and f is twice continuously differentiable.

A central component of our approach involves repeated uniform random subsampling without replacement from P , each of size v . Under the regime where $n \rightarrow \infty$ and $v = o(n)$, the empirical distribution of such a subsample converges weakly to $\mathcal{F}$ [1]. Consequently, for sufficiently large n , each subsample can be treated as approximately i.i.d. from $\mathcal{F}$ in our asymptotic analysis.

The theoretical idea of GBSK rests on three main components, and the details are presented in the following subsections:

- **Rooted in KDE theory.** Kernel density estimation (KDE) provides a classical foundation for mode detection. Under mild regularity conditions, the modes estimated from a single i.i.d. sample converge to the true modes of the underlying density and are asymptotically normal around them. This property holds for both single-mode and well-separated multi-modal densities.
- **Lightweight ordinal proxy.** Pointwise KDE becomes computationally prohibitive for large-scale data. GBSK circumvents this cost by introducing the granular-ball density statistic $\rho_i = n_i/r_i$ as an approximate local proxy for KDE. Rather than estimating exact density values, ρ_i preserves the relative ordering of densities among nearby points—which is sufficient for mode detection, since identifying a mode only requires knowing whether a point outranks its neighbors.
- **Multi-sample refinement.** While we do not claim that ρ_i inherits all asymptotic properties of KDE, our theoretical analysis suggests that it approximates KDE's ordinal behavior locally. Building on this proxy, GBSK performs multiple independent subsampling iterations, each producing a set of candidate modes. And we conjecture that these candidate modes are also asymptotically normal around the true modes. Thus, aggregating them across all iterations (e.g., through a weighted average) further reduces estimation variance and yields progressively more accurate mode locations. This multi-sample refinement strikes a practical balance between statistical fidelity and computational efficiency.

A. Theory of KDE

Let $X = \{x_1, \dots, x_v\} \subset P$ be a subsample of size v drawn uniformly at random without replacement from the full dataset P . Following the asymptotic equivalence discussed previously, when $n = |P|$ is sufficiently large and $v = o(n)$, X can be treated as approximately independently and identically drawn from $\mathcal{F}$ for theoretical analysis.

The kernel density estimator (KDE) for the underlying density $f(x)$ is defined as:

$$\hat{f}_X(x) = \frac{1}{vh_v^d} \sum_{i=1}^v K\left(\frac{x - x_i}{h_v}\right), \quad (22)$$

where $K : \mathbb{R}^d \rightarrow [0, \infty)$ is a kernel function (typically a probability density satisfying $\int_{\mathbb{R}^d} K(\mathbf{z}) d\mathbf{z} = 1$), and $h_v > 0$ is a bandwidth parameter that controls the smoothness of the estimate. For $\hat{f}_X$ to be a consistent estimator of f , the bandwidth sequence must satisfy:

$$h_v \rightarrow 0 \quad \text{and} \quad vh_v^d \rightarrow \infty \quad \text{as} \quad v \rightarrow \infty. \quad (23)$$

A *mode* (or density peak) of f is defined as a strict local maximum point. Denote the set of all modes by $\text{MOD} = \{\text{MOD}_1, \dots, \text{MOD}_{K_0}\}$. Each mode MOD_j satisfies the regularity conditions:

$$\nabla f(\text{MOD}_j) = \mathbf{0} \quad (\text{stationary point}), \quad (24)$$

$$\nabla^2 f(\text{MOD}_j) \quad \text{is negative definite (local maximum)}. \quad (25)$$

Naturally, the local maxima of $\hat{f}_X(\mathbf{x})$ can be used to estimate MOD. Let $\widehat{\text{MOD}}_X$ denote the set of estimated modes from sample X , and $\widehat{\text{MOD}}_X^{(j)}$ be the estimated mode corresponding to the true mode MOD_j .

Lemma 1 (Asymptotic normality of KDE mode estimation (Theorem 2 in [2] and [3])). *Under standard regularity conditions (density smoothness, mode curvature, kernel differentiability, and bandwidth conditions), for a single mode MOD, the normalized estimation error converges to a multivariate normal distribution:*

$$\sqrt{vh_v^{d+2}}(\widehat{\text{MOD}}_X - \text{MOD}) \xrightarrow{d} n(\mathbf{0}, \Sigma), \quad (26)$$

where Σ is covariance matrix.

Lemma 2 (Multi-mode matching and convergence rate (Lemma 3 in [4])). *When the true density has K_0 well-separated modes $\text{MOD}_1, \dots, \text{MOD}_{K_0}$, under appropriate conditions on the bandwidth h_v and sample size v , the estimated modes $\widehat{\text{MOD}}_X$ satisfy:*

- 1) **Existence and uniqueness:** With probability approaching 1, each true mode MOD_j has exactly one estimated mode $\widehat{\text{MOD}}_X^{(j)}$ in its neighborhood.
- 2) **Joint asymptotic normality:** Each $\widehat{\text{MOD}}_X^{(j)}$ is asymptotically normal around MOD_j , with the same covariance structure as in the single-mode case.

We assume the following conditions hold throughout our analysis of multi-modal densities:

- 1) **Mode Separation:** The true modes $\text{MOD}_1, \dots, \text{MOD}_{K_0}$ are well-separated.
- 2) **Local Independence:** For sufficiently small bandwidth h_v , the estimation neighborhoods of different modes do not overlap significantly, making the mode estimation problems asymptotically independent.
- 3) **Regularity Conditions:** Each mode MOD_j satisfies the same regularity conditions as in the single-mode case (Lemma 1), including smoothness of f in its neighborhood.

Under these assumptions, the estimation problem for each mode becomes locally independent in the asymptotic sense. Specifically, for each fixed mode MOD_j , the local estimation

behavior is primarily determined by the following two factors without significant interference from other modes:

- the local curvature of f at MOD_j ,
- the sample information falling within the neighborhood of MOD_j ,

Corollary 1 (Asymptotic normality in the multi-mode setting). *Under the assumptions stated above and with bandwidth h_v chosen appropriately, the asymptotic normality established in Lemma 1 extends to each individual mode. That is, for each $j = 1, \dots, K_0$,*

$$\sqrt{vh_v^{d+2}}(\widehat{\text{MOD}}_X^{(j)} - \text{MOD}_j) \xrightarrow{d} n(\mathbf{0}, \Sigma_j), \quad (27)$$

where Σ_j is defined analogously to Σ in (26), with MOD replaced by MOD_j and the corresponding local properties of f at MOD_j .

Based on Lemma 1, Lemma 2, and Corollary 1, when the sample size is sufficiently large, the estimated mode $\widehat{\text{MOD}}_X^{(j)}$ approximately follows a multivariate normal distribution centered at the true mode MOD_j .

B. Granular-ball Density as a Local Approximate KDE Proxy: An Engineering Perspective

Our method employs a granular-ball density statistic defined in (3) as:

$$\rho_i = \frac{|X_{B_i}|}{r_i} = \frac{n_i}{r_i}, \quad (28)$$

which captures the *mass concentration* at the ball center B_i . c., analogous to how kernel density estimation (KDE) models density via mass accumulation around data points.

It should be noted that ρ_i does *not* coincide with the formal density definition in KDE, where the local density at a point x scales as $\frac{1}{r^d}$ to ensure dimensional consistency in $\mathbb{R}^d$. Nevertheless, under appropriate conditions, ρ can reliably approximate the *ordinal relationships* among density values—which is sufficient for mode identification, the core objective of density-based clustering.

1) *Core Design Philosophy*: Mode identification is inherently local—a point is a mode only if it outranks its neighbors, regardless of distant densities. In a small neighborhood around a mode, the density distribution is similar, which suggests that the exact scaling $1/r^d$ may not be strictly necessary, and the simpler $1/r$ statistic can often preserve the ordinal information needed for local comparison. This property makes the granular-ball density $\rho_i = n_i/r_i$ a viable approximate proxy for KDE in the context of mode detection.

2) *The Balanced Granular-Ball Assumption*: Let n be the total number of data points and M be the number of generated granular-balls. Through our balanced granular-ball generation algorithm (Algorithm 1), we enforce an approximate balancing constraint:

$$n_i = \frac{n}{M} \cdot (1 + \epsilon_i), \quad \text{where } |\epsilon_i| \leq \tau. \quad (29)$$

Here $\tau \in (0, 1)$ is a *balance tolerance* parameter. This constraint has both practical and theoretical motivations:

- **Practical motivation**: It prevents extreme imbalances where some balls contain very few points while others contain many, ensuring stable numerical computation.
- **Theoretical intuition**: In the limit $\tau \rightarrow 0$, we obtain perfectly balanced balls with $n_i = n/M$, which provides the cleanest theoretical analysis.
- **Algorithmic benefit**: The constraint guides the granular-ball generation toward a more regular tessellation of the data space.

3) *From Balance Constraint to Density Approximation*: Under assumption (29), the granular-ball radius r_i becomes a *natural adaptive bandwidth* that reflects local density. Consider the expected number of points in ball B_i :

$$\mathbb{E}[n_i] \approx n \cdot f(c_i) \cdot V_d \cdot r_i^d, \quad (30)$$

where $f(c_i)$ is the true density at the ball center c_i , and V_d is the volume of a unit ball in $\mathbb{R}^d$. Combining with (29):

$$\frac{n}{M} \cdot (1 + \epsilon_i) \approx n \cdot f(c_i) \cdot V_d \cdot r_i^d. \quad (31)$$

Solving for r_i :

$$r_i \approx \left(\frac{1 + \epsilon_i}{M \cdot f(c_i) \cdot V_d} \right)^{\frac{1}{d}}. \quad (32)$$

Now substitute into our density statistic $\rho_i = n_i/r_i$:

$$\rho_i = \frac{n}{M} \cdot (1 + \epsilon_i) \cdot \left(\frac{M \cdot f(c_i) \cdot V_d}{1 + \epsilon_i} \right)^{\frac{1}{d}} \quad (33)$$

$$= \frac{n}{M^{1-\frac{1}{d}}} \cdot V_d^{\frac{1}{d}} \cdot f(c_i)^{\frac{1}{d}} \cdot (1 + \epsilon_i)^{1-\frac{1}{d}}. \quad (34)$$

4) *The Critical Simplification*: The key observation is that for mode detection, we only need to preserve **ordinal relationships**, not exact density values. Let $\mathcal{C} = \frac{n}{M^{1-\frac{1}{d}}} V_d^{\frac{1}{d}}$ be a constant (for fixed n, M, d). Then:

$$\rho_i = \mathcal{C} \cdot f(c_i)^{\frac{1}{d}} \cdot (1 + \epsilon_i)^{1-\frac{1}{d}}. \quad (35)$$

Since the function $g(x) = x^{\frac{1}{d}}$ is **strictly increasing** for $x > 0$, and $(1 + \epsilon_i)^{1-\frac{1}{d}}$ is close to 1 when $|\epsilon_i| \leq \tau \ll 1$, we obtain:

$$\rho_i > \rho_j \iff f(c_i) > f(c_j) \quad \text{with high probability.} \quad (36)$$

The equivalence holds because:

- 1) $g(x) = x^{\frac{1}{d}}$ preserves order (if $f(c_i) > f(c_j)$, then $f(c_i)^{\frac{1}{d}} > f(c_j)^{\frac{1}{d}}$)
- 2) The perturbation factors $(1 + \epsilon_i)^{1-\frac{1}{d}}$ are approximately equal for all balls (bounded by τ)
- 3) For mode detection, we only need *local comparisons* where density contrasts typically exceed the small perturbations introduced by ϵ_i

5) *The Role of τ in Error Analysis*: The tolerance τ controls the approximation quality. Consider two cases:

- **Perfect balance** ($\tau = 0$): All $n_i = n/M$, so $\epsilon_i = 0$. Then:

$$\rho_i = \mathcal{C} \cdot f(c_i)^{\frac{1}{d}}. \quad (37)$$

This gives **approximate ordinal preservation**: $\rho_i > \rho_j \iff f(c_i) > f(c_j)$.

- **Approximate balance** ($\tau > 0$): The factors $(1 + \epsilon_i)^{1-\frac{1}{d}}$ introduce small perturbations.

From the expression ρ_i , the relative deviation from its ideal value $\mathcal{C}f(c_i)^{\frac{1}{d}}$ is:

$$\frac{\rho_i}{\mathcal{C}f(c_i)^{\frac{1}{d}}} - 1 = (1 + \epsilon_i)^{1-\frac{1}{d}} - 1. \quad (38)$$

Since $|\epsilon_i| \leq \tau$ and the function $g(\epsilon) = (1 + \epsilon)^{1-\frac{1}{d}}$ is monotonically increasing for $\epsilon \in [-\tau, \tau]$ (because $1 - \frac{1}{d} > 0$), the maximum absolute deviation occurs at $\epsilon_i = \tau$:

$$\max \left| \frac{\rho_i}{\mathcal{C}f(c_i)^{\frac{1}{d}}} - 1 \right| = (1 + \tau)^{1-\frac{1}{d}} - 1. \quad (39)$$

For small τ , a first-order Taylor expansion of $(1 + \tau)^{1-\frac{1}{d}}$ is:

$$(1 + \tau)^{1-\frac{1}{d}} = 1 + \left(1 - \frac{1}{d}\right) \tau + o(\tau), \quad (40)$$

and consequently,

$$(1 + \tau)^{1-\frac{1}{d}} - 1 \approx \left(1 - \frac{1}{d}\right) \tau. \quad (41)$$

This quantifies the *worst-case relative deviation* between our computed density proxy ρ_i and its ideal monotonic function $\mathcal{C}f(c_i)^{\frac{1}{d}}$.

Adaptive τ Selection: In practice, we recommend setting τ adaptively based on dimension and sample size:

$$\tau = \min \left(0.1, \frac{2.0}{\sqrt{n/M}} \cdot \frac{4}{\max(d, 4)} \right). \quad (42)$$

This formula automatically reduces τ for high-dimensional or small-sample scenarios where approximation errors would be more consequential, while allowing larger τ (and thus faster computation) when the data provides sufficient redundancy.

Error Analysis and Practical Acceptability: For dimensions $d = 2$ to 100, our density proxy exhibits bounded approximation error:

- With $\tau = 0.1$: relative error $\in [4.88\%, 15.14\%]$;
- With $\tau = 0.2$: relative error $\in [9.54\%, 33.99\%]$.

In the worst case, e.g., $d = 100$, the error remains below 15.2% with $\tau = 0.1$, and the average error is about 8.5%.

Remark: As mentioned above, mode identification is inherently local. Furthermore, since the pdf $f(x)$ is twice continuously differentiable, within a sufficiently small neighborhood, the closer two granular-balls are, the more similar their local density distributions become, and hence the more similar their approximation errors in relative ordering. This natural correlation ensures that the comparisons which actually determine mode assignments are also more reliable, and the mode candidate identification in Algorithm 3 inherently matches this principle. It utilizes criterion of the density peaks—points with high local density and large distance to the nearest higher-density neighbor—by design selects local maxima based solely on comparisons with spatial neighbors, filtering distant points. Thus, the comparisons that matter most for mode detection

tend to be confined to nearby points, where our density proxy tends to be more accurate.

The analysis and discussion above demonstrate that $\rho_i = \frac{n_i}{r_i}$, under the balanced granular-ball constraint, is approximately and locally proportional to $f(c_i)^{\frac{1}{d}}$. Combined with the local nature of mode detection, this guarantees that the ordinal information essential for clustering is preserved with bounded, practically acceptable error. Empirical results (Section V) confirm that these theoretical bounds are conservative and that GBSK achieves accuracy comparable to exact KDE-based methods.

Consequently, the granular-ball density statistic serves as a theoretically grounded, computationally efficient approximate proxy for kernel density estimation. This interpretation aligns with classical results on distance-based density proxies (e.g., the kNN estimator $\hat{f}_{\text{kNN}}(x) \propto 1/R_k(x)^d$, where $R_k(x)$ is the distance to the k -th nearest neighbor [5]), and leads to the following conjecture about asymptotic equivalence.

Conjecture 1 (Structural Equivalence of Mode Estimation). *Under the conditions established in Lemma 1, Lemma 2, Corollary 1 and the approximate ordinal preservation property in (36), the following holds:*

Let $\{B_i\}_{i=1}^M$ be a set of granular-balls generated from a subsample X of size v , with density statistics $\rho_i = n_i/r_i$. Suppose B_j is a granular-ball whose center c_j is a local maximum of $\{\rho_i\}$ in its neighborhood.

Let $\hat{f}_X(x)$ be the KDE estimator defined in (22), and let x^ be a local maximum of $\hat{f}_X(x)$ within an ϵ -neighborhood $\mathcal{N}_\epsilon(x) = \{y \mid \|x - y\| < \epsilon\}$.*

Then, as $v \rightarrow \infty$, $h_v \rightarrow 0$, $vh_v^d \rightarrow \infty$, and $\epsilon \rightarrow 0$ appropriately, both c_j and x^ converge in probability to the same population mode MOD_j :*

$$c_j \xrightarrow{P} \text{MOD}_j, \quad x^* \xrightarrow{P} \text{MOD}_j. \quad (43)$$

Consequently, mode identification via granular-ball density peaks and via KDE gradient ascent are asymptotically consistent, both recovering the true set of population modes $\{\text{MOD}_1, \dots, \text{MOD}_{K_0}\}$.

C. Theoretical Validity of GBSK

Building on the approximate local ordinal preservation property established in (36) and the conjectured asymptotic equivalence to KDE mode detection, we now examine the theoretical underpinnings of the GBSK algorithm.

- 1) **Independent Resampling:** From the sufficiently large dataset P , perform s independent resampling iterations. In each iteration i , draw a subsample of size v :

$$P^{(i)} = \{p_{i1}, \dots, p_{iv}\}, \quad p_{ij} \stackrel{\text{i.i.d.}}{\sim} \mathcal{F}. \quad (44)$$

- 2) **Approximate local KDE proxy via Granular-balls:** For each subsample $P^{(i)}$, construct granular-balls and compute the granular-ball density statistic ρ , which serves as a computationally efficient approximate proxy for kernel density estimation.
- 3) **Representative Point Extraction and Aggregation:** Using DPeak [6], extract the top k local density peaks

(i.e., local modes) from the granular-ball representation, each of which has relatively high density within its neighborhood and is sufficiently distant from points of higher density. Then, combine all of them form the aggregated set (Section IV Step 2):

$$ARBC = \{repBallCenters_j^{(i)}\}_{s \times k}. \quad (45)$$

where $repBallCenters_j^{(i)}$ is an estimation of MOD_j in sample set $P^{(i)}$.

- 4) **Final Mode Estimation:** Perform a second-stage granular-ball density estimation on the aggregated set $ARBC$ to estimate the final modes of the underlying distribution.

From the above theory and analysis, the estimation of the mode based on granular-ball density statistics can serve as an approximate local proxy for the estimation of kernel density, each representative point $repBallCenters_j^{(i)}$ can be regarded as a sample drawn from the asymptotic distribution of the corresponding true mode MOD_j .

By Lemma 2, when the subsample size v is sufficiently large, with high probability there exists a bijection between the estimated modes $\{repBallCenters_j^{(i)}\}_{j=1}^{K_0}$ and the true modes $\{MOD_j\}_{j=1}^{K_0}$, such that each $repBallCenters_j^{(i)}$ lies in a neighborhood of its corresponding MOD_j . Furthermore, Corollary 1 establishes that for each true mode MOD_j , the corresponding estimated mode satisfies asymptotic normality:

$$\sqrt{vh_v^{d+2}}(repBallCenters_j^{(i)} - MOD_j) \xrightarrow{d} n(\mathbf{0}, \Sigma_j), \quad (46)$$

where Σ_j is the covariance matrix determined by the local curvature of f at MOD_j and the kernel function.

Since the resampling iterations are mutually independent, for each fixed j , the sequence $\{repBallCenters_j^{(i)}\}_{i=1}^s$ consists of independent and identically distributed asymptotically normal samples.

Let $ARBC_j = \{repBallCenters_j^{(i)}\}_{i=1}^s$ denote the set of representative points corresponding to the true mode MOD_j collected from all s resampling iterations.

When the number of resampling iterations $s \rightarrow \infty$, the following distributional properties emerge:

- 1) **Clustering Structure:** With high probability, all representative points in $ARBC_j$ form a compact cluster, while clusters corresponding to distinct modes remain well-separated. Provided that the number of extracted peaks per subsample satisfies $k \geq K_0$, the set $ARBC$ naturally partitions into K_0 distinct clusters, each associated with one true mode.
- 2) **Center Convergence:** The sample mean of $ARBC_j$,

$$\overline{ARBC}_j = \frac{1}{s} \sum_{i=1}^s repBallCenters_j^{(i)}, \quad (47)$$

is a refined estimator of MOD_j , converging in probability as $s \rightarrow \infty$.

- 3) **Normal Distribution:** Within each $ARBC_j$, the representative points asymptotically follow a multivariate normal distribution centered at MOD_j with covariance

scaling as $O(1/(vh_v^{d+2}))$, where h_v is the bandwidth used in the KDE-based mode estimation (see Lemma 1).

These distributional properties demonstrate that when the number of resampling iterations s is sufficiently large, the representative point set $ARBC$ densely populates the neighborhoods of the true modes and forms approximately Gaussian scatter around each mode. This provides the theoretical foundation for the final stage of GBSK: performing secondary density estimation on $ARBC$ to extract the final modes. The distributional structure of $ARBC$ inherently encodes the positions of the true modes, enabling accurate recovery through appropriate density estimation methods.

APPENDIX II

DETAILS OF NUMERICAL DATASETS

Dataset	Instances	Dims	Classes	Type	File Size	Usage	Label	Description
CHAINLINK	1,000	3	2	S	20.6KB	Q	Y	The points are arranged into two interlocking 3D rings. Non-linearly separable data. https://github.com/milaan9/Clustering-Datasets/blob/master/02.%20Synthetic/chainlink.csv
TWENTY	1,000	2	20	S	17KB	Q	Y	The dataset contains 20 evenly distributed, distinct clusters, ideal for evaluating clustering algorithms on separable data. https://github.com/milaan9/Clustering-Datasets/blob/master/02.%20Synthetic/twenty.mat
SYN1 [7]	2,000	2	5	S	30.1KB	Q	Y	Each cluster varies in size and shape, demonstrating different densities and distributions.
SEGMENTATION	2,310	18	7	R	246KB	Q	Y	Derived from hand-segmented outdoor images for classification tasks. https://doi.org/10.24432/C5P01G
SYN3 [7]	3,000	2	2	S	3.5KB	Q	Y	The clusters exhibit different sizes and densities, including one non-convex cluster.
ENGYTIME	4,096	2	2	S	76.5 KB	Q	Y	Derived from two Gaussian distributions, with overlapping classes and a parabolic decision boundary. https://github.com/milaan9/Clustering-Datasets/blob/master/02.%20Synthetic/engytime.arff
WAVEFORM	5,000	21	3	R	520KB	Q	Y	Designed for classification tasks in physics and chemistry. The features simulate combinations of three basic waveforms with added noise. https://doi.org/10.24432/C5CS3C
S3	5,000	2	15	S	136KB	Q	Y	S3 contains 5,000 vectors divided into 15 Gaussian clusters with varying degrees of overlap. https://cs.joensuu.fi/sipu/datasets/s3.txt
SYN2 [7]	5,800	2	7	S	7.1KB	Q	Y	Diverse cluster shapes including a non-convex cluster.
PENDIGITS	10,992	16	10	R	685KB	Q, S	Y	Pen-based handwritten digits (0-9) formatted as 16-dimensional feature vectors for pattern recognition. https://doi.org/10.24432/C5MG6K
DRYBEAN	13,611	16	7	R	3.6MB	Q, S	Y	It consists of dry beans from 7 types, characterized by 16 geometric features derived from digital images to aid in classification tasks. https://doi.org/10.24432/C50S4B
CIFAR-10	60,000	3,072	10	R	174MB	Q, S	Y	Consists of 32x32 colour images in 10 classes, with 6000 images per class. https://api.semanticscholar.org/CorpusID:18268744
MNIST	70,000	784	10	R	121MB	Q, S	Y	A widely used benchmark in machine learning, consisting of grayscale images of handwritten digits from 0 to 9. https://www.csie.ntu.edu.tw/~cjlin/libsvmtools/datasets/multiclass.html#mnist
MoCAP	78,095	36	5	R	43MB	Q, S	Y	5 types of hand postures from 12 users were recorded using unlabeled markers attached to fingers of a glove in a motion capture environment. https://doi.org/10.24432/C5960R
RCV1	534,135	47,236	53	R	fileSize	Q, S	Y	A large-scale text dataset derived from the Reuters Corpus Volume I, represented by sparse TF-IDF features after standard preprocessing, with category labels mapped to the second-level topic hierarchy. https://www.csie.ntu.edu.tw/~cjlin/libsvmtools/datasets/multiclass.html#rcv1.multiclass
COVERTYPE	581,012	54	7	R	71.1MB	Q, S	Y	Forest cover type samples from US Forest Service Region 2 Resource Information System data, with 54 features including soil type and wilderness area. https://doi.org/10.24432/C50K5N
3M2D5 [8]	3,000,000	2	5	S	53.7MB	Q, S	Y	A mixture of 5 bivariate normal distributions with specified means, covariances, and mixing proportions, forming visually apparent but mixed clusters.
N-BaIoT	7,062,606	115	9	R	7.6GB	S	N	Real traffic data from nine commercial IoT devices, authentically infected by Mirai and BASHLITE malware. A valuable resource for researching IoT security and anomaly detection methodologies. https://www.kaggle.com/datasets/mkashifn/nbaiot-dataset
MNIST8M	8,100,000	784	10	R	5.9GB	Q, S	Y	An augmented version of the MNIST training set, created using elastic deformation to generate variations of handwritten digits. https://www.csie.ntu.edu.tw/~cjlin/libsvmtools/datasets/multiclass.html#mnist8m
AGC100M	100,000,000	256	17	S	95.3GB	Q, S	Y	AGC100M, with its 17 clusters of varying sizes and shapes, complex anisotropic structures, and 5% noise, is optimally designed for robustness testing of clustering algorithms in high-dimensional settings. https://www.kaggle.com/datasets/caatic7/agc100m

“Type”: “S” denotes synthetic datasets, “R” means real datasets.

“Usage”: “Q” indicates clustering quality evaluation, “S” for clustering speed.

“Label”: “Y” indicates labeled data, “N” for unlabeled.

APPENDIX III EXTENDED SCALABILITY NOTES

Parameter Configuration. To ensure a fair comparison, competitor parameters were optimized according to their respective literature recommendations.

- **AGBSK and k-means++:** Configured solely with the target cluster count k .
- **FastDPeak, FHC-LDP, Fast LDP-MST:** Configured with the number of nearest neighbors K (for density estimation) and cluster count. For brevity, Table II reports only the sensitivity parameter K , with cluster count held constant across methods.
- **GB-DP:** Operates as a parameter-free method up to the decision graph stage, where cluster centers are selected manually. The reported runtimes for GB-DP exclude this manual interaction time.

Dataset-Specific Runtime Notes.

- **Moderate-scale** (PENDIGITS, DRYBEAN, MOCAP): GBSK matches baseline runtimes except for k-means++ while achieving equal or higher ACC (Table II); the overhead of the multi-sampling strategy is negligible compared to the gains in structural precision.
- **Large-scale** (COVERTYPE, MNIST, CIFAR-10): As dataset cardinality increases, the computational limitations of competitor methods become acute. Algorithms relying on global kNN graphs or full-matrix operations (FastDPeak, FHC-LDP, Fast LDP-MST) succumb to exponential time complexity or memory exhaustion; GBSK/AGBSK retain near-linear growth by processing sampled subsets independently.
- **Web-scale** (MNIST8M, AGC100M): only the proposed methods terminate within 2000 s; AGBSK complete AGC100M with peak memory below 12 GB. The divergence point at roughly 6×10^6 samples in Fig. 6 marks the onset of GBSK’s sustained throughput advantage.

APPENDIX IV SCALABILITY IMPLEMENTATION FOR AGC100M

To process the largest dataset AGC100M within the memory constraints of a standard workstation, we implemented a block-wise, out-of-core processing strategy. This approach effectively treats GBSK as a two-pass streaming algorithm:

- 1) **Pass 1: Streaming Sampling.** The raw data file is read in $m = 100$ sequential batches to minimize peak memory usage. From each batch, we perform reservoir sampling to select $\frac{\sqrt{n}}{m}$ points. These subsamples are aggregated in memory to form the input sets $P^{(1)}, \dots, P^{(s)}$, maintaining the global sampling ratio of $\alpha \approx n^{-1/2}$ consistent with the AGBSK default.
- 2) **In-Memory Skeleton Construction.** Once the representative samples are loaded, Steps 2–4 of the GBSK protocol (Representative Ball selection, Key Ball refinement, and Skeleton construction) are executed entirely in memory. This phase requires negligible RAM compared to the full dataset size.
- 3) **Pass 2: Streaming Assignment.** To generate the final cluster labels (Step 5), the dataset is streamed a second time in batches. For each incoming block of data, points are assigned to the nearest cluster in the pre-computed skeleton, and the results are written directly to disk.

The runtime of each step for the proposed GBSK method on AGC100M dataset is provided in Fig. 15 and Table XII–XIII.

APPENDIX V RESOURCE UTILIZATION ANALYSIS

To validate the practical deployability of GBSK on standard workstations, we monitored the system resource consumption during the processing of the AGC100M dataset (95.3 GB binary size). Performance metrics were logged at one-second intervals using the Windows Performance Monitor; the complete trace is available at https://github.com/XFastDataLab/GBSK/blob/main/experiment_records/MATLAB_Collector%20PerfMon.csv.

A. Memory Efficiency via Streaming

The peak memory footprint of MATLAB running GBSK/AGBSK remained consistently below **12 GB**, contrasting sharply with the 95.3 GB required to hold the full dataset in memory. This represents an approximate **88% reduction** in peak memory demand. This efficiency is directly attributable to the streaming architecture of the sampling (Step 1) and assignment (Step 5) phases. By loading data in sequential chunks rather than all data, the algorithm decouples the maximum memory requirement from the total dataset size.

- **Peak Memory Event:** The highest memory usage (11.62 GB) occurred during the final assignment step, where data chunks are temporarily loaded to compute distances against the skeleton.
- **Comparison with Baselines:** Standard implementations of k-means++ and GB-DP mandate loading the entire dataset into RAM. Consequently, these methods failed to initialize on AGC100M and MNIST8M due to Out-Of-Memory errors, highlighting GBSK as the only viable option for datasets exceeding available physical memory.

B. CPU Utilization and Precision

The peak CPU utilization remained below **70%**, indicating that the algorithm is not bottlenecked by processor saturation even during intensive distance computations.

Note on Numerical Precision: Unless otherwise stated, all experiments utilized MATLAB’s default double-precision floating-point format. However, for the AGC100M dataset, to further optimize throughput and storage, both data storage and internal computations were performed using single-precision floating-point arithmetic.

APPENDIX VI DISCUSSION

A. Behavior in High-Dimensional Spaces

In Table VI, the result on RCV1 text corpus reveals a distinct behavioral dichotomy in GBSK as the ambient dimensionality increases. Specifically, we observe that while the *probability of successfully initializing* a valid skeleton decreases with dimension, the *topological fidelity* of the skeleton, once obtained, remains remarkably stable. This phenomenon suggests that the primary bottleneck in high-dimensional settings is *structural emergence* rather than *structural exploitation*. In high-dimensional spaces, the concentration of measure phenomenon causes pairwise distances to converge, weakening local density contrasts. This flattening of the density landscape makes it increasingly difficult to reliably identify compact, high-density granular-balls during the initial sampling phase. Consequently, the algorithm may terminate early if sufficiently dense anchors cannot be robustly detected.

Crucially, however, the subsequent stages of GBSK (aggregation and assignment) are largely invariant to dimensionality. Once the geometric density skeleton is established, it effectively serves as a low-dimensional manifold approximation, shielding the final assignment steps from the noise of the high-dimensional ambient space. Methodologically, GBSK prioritizes structural consistency over forced convergence: it refuses to produce a clustering result if the density signal is too weak, rather than returning a spurious partition. In practice, this initialization sensitivity is effectively mitigated by the multi-sampling strategy, where independent restarts substantially increase the likelihood of recovering the true density ridge [9] without compromising the quality of the result.

B. Robustness to Noise and Outliers

Robustness constitutes a fundamental prerequisite for clustering algorithms deployed in real-world scenarios, where measurement errors, rare events, and spurious samples are inevitable. As consistently observed across complex benchmarks, GBSK demonstrates intrinsic resilience to such perturbations. This capability is underpinned by three synergistic mechanisms: granular smoothing, ensemble variance reduction, and skeleton-based topology recovery.

Implicit Regularization via Granular Smoothing. The primary source of robustness in GBSK lies in its granular-ball representation. By operating on aggregated hyperspheres rather than individual data points, which are highly susceptible to local variance, GBSK implicitly applies a geometric *low-pass filter* to the data distribution. In this framework, isolated noise points and small outlier groups typically fail to amass sufficient local density to trigger the selection of a representative ball. Furthermore, even if such balls are initially formed, they are effectively discarded during the *Key Ball* generation stage due to their instability. Consequently, the constructed skeleton represents only the persistent high-density backbone of the data, filtering out high-frequency noise components.

Variance Reduction via Multi-Sampling. The multi-sampling strategy further fortifies robustness by mitigating the

TABLE VI: AGBSK performance on RCV1 (of 53 classes: second level of RCV1 topic hierarchy) with varying feature dimensions. Valid Ratio denotes the proportion of successful runs out of 100 total runs.

d	Valid Ratio	Time (s)	#ARBC	#Key Balls	ACC
128	91%	6.3±0.2	1143.7±22.1	232.9±9.2	0.311±0.005
512	91%	9.5±0.4	833.5±30.3	120.2±7.5	0.317±0.005
1024	68%	19.1±0.9	867.9±23.8	99.1±7.7	0.322±0.008
2048	44%	42.3±2.4	982.7±42.2	82.4±6.8	0.327±0.010

stochastic bias of any single subsample. Under the ensemble learning paradigm, the aggregation of skeletons from multiple independent subsamples serves as a variance reduction mechanism. While noisy or outlying samples may appear sporadically in individual subsamples, they lack the persistence to form stable structural components across multiple subsamples. In contrast, regions corresponding to the true underlying manifold are consistently captured. This consensus-based aggregation ensures that the final skeleton reflects the intrinsic, persistent geometry of the data distribution rather than transient sampling artifacts.

Topological Stability via Manifold Learning. From a manifold learning perspective, GBSK prioritizes *topology recovery* over the precise delineation of cluster boundaries. Since noise and outliers predominately reside in the low-density voids between clusters (the valleys of the probability density function), they are geometrically decoupled from the skeleton formation process. This design choice ensures that the global clustering structure remains stable even in the presence of significant local perturbations, as the skeleton anchors itself to the density peaks rather than the unstable peripheral boundaries.

Limitations and Failure Modes. Notwithstanding these strengths, specific pathological noise conditions can still compromise the performance of framework:

- 1) *Structured Outliers*: When noise is not random but structured—forming dense, cohesive groups comparable to genuine clusters—it may generate spurious granular-balls that persist across sampling rounds, leading to phantom skeleton branches.
- 2) *Heterogeneous Noise Distribution*: If noise is heavily concentrated within specific clusters, it may distort local density estimation, potentially causing skeleton fragmentation in high-dimensional spaces.
- 3) *High-Density Background Noise*: In scenarios where a large proportion of samples constitute uniform background noise, the density contrast between the manifold core and the background diminishes. This blurring complicates the adaptive estimation of granular-ball radii, potentially merging distinct clusters.

These limitations reflect a fundamental challenge in density-based clustering: distinguishing signal from noise becomes inherently ambiguous when the noise itself exhibits structural characteristics indistinguishable from the signal.

C. Robustness to Cluster Imbalance

This subsection provides a critical analysis of the impact of cluster imbalance on the GBSK framework. Real-world data

distributions frequently exhibit significant disparities in cluster cardinality, density, and spatial volume, posing a challenge for algorithms that assume balanced priors.

Mechanisms of Sensitivity. the reliance of GBSK on granular abstraction introduces specific behaviors under imbalance. Large, high-density clusters naturally generate a rich set of stable granular-balls, facilitating the construction of robust, continuous skeleton structures. Conversely, minority clusters, characterized by small sample sizes or low relative densities, may be represented by a sparse set of granular-balls.

- **Fragmentation Risk**: If the local density of a minority cluster falls below the adaptive selection threshold used during Key Ball generation, its skeleton may become fragmented or, in extreme cases, fail to emerge entirely.
- **Resolution Limit**: The granular-ball count M implicitly sets a minimum resolution; extremely small clusters may be treated as noise if their volume is insufficient to support a distinct granular-ball.

Intrinsic Advantages over Partitioning Methods. Despite these constraints, GBSK demonstrates superior resilience to size imbalance compared to classical distance-based partitioning methods (e.g., k-means).

- **Local Density Autonomy**: Unlike centroid-based algorithms, which are biased toward creating clusters of equal variance (often splitting large clusters or merging small ones), the skeleton extraction of GBSK is governed strictly by *local density ordering*.
- **Decoupled Analysis**: Dominant clusters do not exert a global gravitational pull that absorbs smaller neighbors. Instead, each cluster is resolved within its own local density manifold. Consequently, provided a minority cluster maintains sufficient internal density contrast against its immediate background, its structural identity is preserved regardless of the global imbalance ratio.

D. Comparative Analysis with Graph-Based Pipeline

This subsection presents a detailed empirical comparison between GBSK and the widely adopted FAISS [10] kNN + Leiden [11] clustering pipeline. As a state-of-the-art method for large-scale unsupervised learning, the combination of FAISS for efficient approximate nearest neighbor search and the Leiden algorithm for community detection represents a rigorous baseline for scalability performance.

Computational Bottlenecks. Quantitative results (Table VII) reveal that the graph construction phase constitutes the primary computational bottleneck. On high-dimensional datasets such as RCV1, the construction of the kNN graph dominates the total runtime. As the ambient dimensionality increases, both execution time and memory consumption exhibit rapid growth. Crucially, on extremely large-scale benchmarks (e.g., MNIST8M), the explicit materialization of the neighbor graph approaches computational intractability, often exceeding feasible time or memory budgets on standard hardware.

Topological Granularity Mismatch. Beyond computational costs, a fundamental divergence is observed in the clustering topology. The FAISS kNN + Leiden pipeline consistently yields a number of clusters that exceeds the underlying

TABLE VII: FAISS kNN + Leiden clustering performance on various datasets.

Dataset	#Clusters	kNN Time (s)	Leiden Time (s)	Mem Peak (GB)
MNIST	267	18.6	7.7	0.9
CIFAR-10	25,797	47.3	2.4	2.5
RCV1_128D	9,809	0.0034	19.5	2.3
RCV1_512D	9,791	682.9	59.5	7.6
RCV1_1024D	9,777	1,474.8	58.5	10.7
RCV1_2048D	9,789	2,646.0	59.6	20.9
MNIST8M	–	timeout	–	–

semantic classes by orders of magnitude. For instance, on the CIFAR-10 dataset, the graph-based approach partitions the data into tens of thousands of fine-grained communities. This phenomenon reflects the intrinsic objective of modularity-based community detection, which tends to over-segment data into micro-communities rather than identifying global modes. This behavior stands in sharp contrast to GBSK, which prioritizes a compact *skeleton representation*. Driven by density structure rather than graph partitioning, GBSK successfully recovers a parsimonious set of clusters that aligns more closely with the semantic categories of the data.

E. Comparative Analysis with Deep Clustering Paradigms

This subsection provides a critical comparative analysis between GBSK and representative deep clustering paradigms. Recent advancements, such as IDEC [12], SDCN [13], Contrastive Clustering (CC) [14], and ProPos [15], have established new benchmarks by unifying feature representation learning and cluster assignment within joint optimization frameworks. While effective, these methods introduce distinct trade-offs regarding computational efficiency, domain universality, and model interpretability.

Computational Efficiency and Training Overhead. A fundamental distinction lies in the computational lifecycle. Deep clustering models necessitate an extensive training phase involving iterative backpropagation, often requiring orders of magnitude more runtime than traditional clustering algorithms. As evidenced in Table VIII, while the inference time (clustering time) of trained models may be reasonable, the end-to-end latency including training is substantial (e.g., 33.5 hours for CC on MNIST). Furthermore, these methods typically depend on specialized hardware (GPUs) and extensive hyperparameter tuning. In contrast, GBSK is a training-free, single-pass framework that extracts structural skeletons directly from the static feature space, enabling efficient processing on standard CPUs with negligible setup costs.

Domain Specificity and Inductive Biases. Many state-of-the-art deep clustering methods exhibit strong modality-specific inductive biases. For instance, contrastive frameworks (e.g., CC, ProPos) rely heavily on data augmentation strategies that are inherently designed for image data. Transferring these paradigms to other domains, such as text (e.g., RCV1) or generic high-dimensional vectors, requires non-trivial architectural redesign to define appropriate views for contrastive learning. GBSK, adopting a density-driven geometric abstraction, is agnostic to the data modality, making it readily appli-

TABLE VIII: Performance and Runtime Comparison: AGBSK vs. Deep Clustering Baselines

Dataset	Metric	AGBSK	IDEC	SDCN	CC	ProPos
PENDIGITS	ACC	0.800	–	–	0.735	0.653
	Train Time	–	–	–	28.8 m	–
	Infer Time	1.5 s	–	–	0.6 s	1.1 h
MNIST	ACC	0.557	0.881	–	0.809	0.830
	Train Time	–	N/A [†]	–	33.5 h	–
	Infer Time	6.8 s	489.0 s	–	25.9 s	5.8 h
RCV1	ACC	0.311	–	0.256	– [‡]	– [‡]
	Train Time	–	–	2.2 h	–	–
	Infer Time	6.3 s	–	0.2 s	–	–

Notes: Time is reported in seconds (s), minutes (m), or hours (h).

[†] IDEC training time on MNIST is N/A as pretrained weights were used.

[‡] CC and ProPos are designed for image augmentation and are not applicable to sparse text data.

cable to heterogeneous datasets where such domain-specific assumptions do not hold.

Interpretability and Structural Transparency. Deep clustering models often operate as black boxes, where cluster assignments are determined by opaque latent space embeddings. Analyzing failure cases (e.g., why two clusters were merged) is challenging due to the entanglement of feature learning and clustering objectives. Conversely, GBSK provides explicit intermediate structures, i.e., Key Balls and Skeletons, that offer a transparent view of the data’s topology. Failure modes in GBSK can be directly traced to geometric causes, such as insufficient density support in specific regions or fragmented skeleton connectivity, facilitating actionable post-hoc analysis.

The choice between deep clustering and GBSK is dictated by the availability of resources and domain constraints. Deep clustering is superior in scenarios where raw data requires complex feature extraction (e.g., raw pixels) and computational resources are abundant. GBSK serves as a robust alternative for general-purpose data mining, prioritizing scalability, interpretability, and cross-domain flexibility.

F. Implications and Limitations

The experimental evidence positions GBSK as a highly effective method for large-scale data mining, particularly when the intrinsic dimensionality of the manifold is moderate ($d \lesssim 1000$). The ability to recover density ridges [9] without full density estimation implies that *explicit structure extraction* is a viable competitor to both implicit density-based methods and deep clustering approaches. The multi-stage abstraction mechanism, i.e., transforming points to balls, balls to skeletons, and skeletons to clusters, provides a scalable blueprint for non-iterative structure learning. This makes GBSK an ideal candidate for exploratory data analysis or as a high-speed initialization module for downstream tasks.

There are some certain limitations in the GBSK method:

- **Topology Mismatch:** The most notable failure mode is incorrect intra-cluster connectivity (Fig. 11), where a Key Ball inadvertently links to a neighbor in a different semantic cluster rather than its true parent. This usually occurs in regions where density ridges [9] are highly

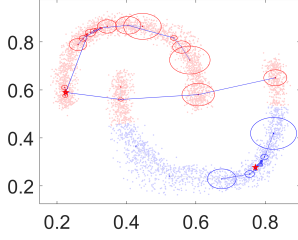


Fig. 11: False skeleton of intra-cluster connections.

anisotropic or where clusters are bridged by high-density noise.

- **Accumulated Approximation Error:** As a multi-stage approximation method, geometric errors at the initial granular-ball level can propagate to the geometric density skeleton. If the initial granular-balls fail to capture the local curvature of the manifold, the aggregated skeleton may short-circuit the true topology.

Future work will focus on mitigating these topological errors, potentially by incorporating local covariance information into the granular-ball distance metric (Mahalanobis distance) or by introducing probabilistic linkage criteria to robustly handle ambiguous cluster boundaries.

G. Skeleton Evaluation

To quantitatively assess the fidelity of the learned skeleton, we evaluate how well it preserves both the global distributional structure and the spatial concentration properties of the original data.

Given a dataset $X = \{x_i\}_{i=1}^n \subset \mathbb{R}^d$ and the learned skeleton composed of density-weighted key balls, we generate synthetic samples $Y = \{y_j\}_{j=1}^m$ from the skeleton by sampling key balls proportionally to their estimated density and structural importance within the skeleton tree.

We then compare X and Y using two complementary metrics: Maximum Mean Discrepancy (MMD) and Skeleton Deviation.

a) Maximum Mean Discrepancy (MMD): MMD measures the discrepancy between two probability distributions via their mean embeddings in a reproducing kernel Hilbert space (RKHS) [16]. The squared MMD between distributions P and Q is defined as

$$\text{MMD}^2(P, Q) = \|\mathbb{E}_{x \sim P}[\phi(x)] - \mathbb{E}_{y \sim Q}[\phi(y)]\|_{\mathcal{H}}^2, \quad (48)$$

where $\phi(\cdot)$ is the feature map associated with a positive definite kernel $k(\cdot, \cdot)$.

Its empirical estimator is

$$\begin{aligned} \widehat{\text{MMD}}^2(X, Y) &= \frac{1}{n^2} \sum_{i, i'} k(x_i, x_{i'}) + \frac{1}{m^2} \sum_{j, j'} k(y_j, y_{j'}) \\ &\quad - \frac{2}{nm} \sum_{i, j} k(x_i, y_j). \end{aligned} \quad (49)$$

We use a Gaussian kernel. A smaller MMD indicates stronger global distributional consistency between the skeleton-induced distribution and the original data distribution.

TABLE IX: Skeleton evaluation results of AGBSK across datasets. Lower values indicate better global alignment (MMD) and stronger spatial adherence (Skeleton Deviation).

Dataset	MMD	Mean Dev.	Median Dev.	Tail Ratio
CIFAR-10	0.1332	2557.9	2539.6	0.0284
MNIST	0.0816	1395.0	1424.6	0.0107
MoCAP	0.0123	7691.5	7806.9	0.0131
PENDIGITS	0.0064	38.30	34.06	0.0438
DRYBEAN	0.0503	150.9	75.9	0.0312

b) Skeleton Deviation: To measure spatial adherence of the skeleton to dominant data regions, we define Skeleton Deviation as the distance from each data point to its nearest skeleton segment.

Let $\mathcal{S}$ denote the set of line segments induced by the skeleton trees. For each $x_i \in X$, we compute

$$d_i = \min_{s \in \mathcal{S}} \text{dist}(x_i, s), \quad (50)$$

where $\text{dist}(x_i, s)$ is the Euclidean distance between x_i and segment s .

We summarize $\{d_i\}$ using:

- Mean deviation: $\frac{1}{n} \sum_{i=1}^n d_i$,
- Median deviation,
- Tail ratio: $\frac{1}{n} \sum_{i=1}^n \mathbf{1}(d_i > \tau)$,

where τ is a dataset-dependent high-percentile threshold.

Smaller values indicate that the skeleton lies closer to the dominant data regions.

c) Discussion: Across datasets (Table IX), lower MMD values generally coincide with smaller tail ratios, indicating that improved global distributional alignment tends to accompany stronger spatial concentration around the skeleton. Low-dimensional structured datasets such as PENDIGITS achieve extremely small MMD values, suggesting effective manifold capture. Although absolute deviation magnitudes vary due to scale differences, the consistently small tail ratios indicate that most data points remain close to the learned skeleton structure.

Together, MMD and Skeleton Deviation provide complementary and interpretable measures for skeleton fidelity in high-dimensional datasets.

H. Local Geometric Refinement via Binary Quantization

A potential concern in high-dimensional clustering is the validity of Euclidean distance. We clarify that the utilization of k-means++ (with $k = 2$) within our framework is *not* intended to identify semantically meaningful clusters at the global scale. Rather, it functions as a local binary quantization operator restricted to the bounded domain of an existing granular-ball.

Within the spatially constrained envelop of a granular-ball, the “curse of dimensionality” is less pronounced compared to the global space. The binary split serves strictly as a geometric refinement step, orienting the partition along the principal axis of local variance. Consequently, the utility of a split is evaluated solely by its ability to reduce the spatial extent of the data representation, not by the semantic correctness of the assignment.

I. Density Ties

A potential limitation of GBSK is the existence of density plateaus, which may make the nearest-higher-density connection ambiguous. In our implementation, we therefore instrumented the code to record every tie encountered during the nearest-higher-density search. Here, a tie is counted whenever two candidate neighbors produce identical density values under either exact comparison or a numerical tolerance of 10^{-12} .

Table X summarizes the tie statistics on the nine large-scale benchmarks (Table II) used in this study. Each dataset was evaluated with 10 independent seeds. As shown in the table, we observed *zero* tie events for all datasets under both exact comparison and tolerance-based comparison. Consequently, the empirical tie frequency is effectively 0% across all benchmarks. This indicates that, in practice, the granular-ball density statistic $\rho = N/r$ produces a sufficiently continuous-valued signal, and exact numerical ties are extremely unlikely in large-scale data.

TABLE X: Tie-counter summary over 10 seeds per dataset.

Dataset	n	Tie events (exact)	Tie events (tol= 10^{-12})	Decision events
Pendigits	1.1×10^4	0	0	929.0
DryBean	1.4×10^4	0	0	1316.2
CIFAR-10	6.0×10^4	0	0	1129.5
MNIST	7.0×10^4	0	0	2091.5
MoCap	7.8×10^4	0	0	1750.0
CoverType	5.8×10^5	0	0	3305.3
3M2D5	3.0×10^6	0	0	1900.4
MNIST8M	8.1×10^6	0	0	18901.9
AGC100M	1.0×10^8	0	0	42233.2

We further examined the run-to-run stability of clustering accuracy by post-processing the 50-run logs already collected for these benchmarks, with each run storing its random seed. The resulting mean ACC and sample standard deviation are reported in Table XI. The average ACC standard deviation across the nine datasets is 0.032, indicating that the tie-breaking heuristic does not introduce measurable variability in the final clustering accuracy.

TABLE XI: ACC mean and standard deviation over the archived 50-run logs.

Dataset	ACC mean	ACC std
Pendigits	0.698	0.050
DryBean	0.531	0.047
MoCap	0.542	0.051
CoverType	0.314	0.040
3M2D5	0.991	0.001
MNIST	0.459	0.043
CIFAR-10	0.191	0.010
MNIST8M	0.389	0.043
AGC100M	0.951	0.004
Average	0.563	0.032

Overall, these results show that although density plateaus are a valid theoretical concern, they are not observed in practice on the evaluated large-scale benchmarks. The empirical tie frequency is zero, and the clustering accuracy remains stable across repeated runs.

APPENDIX VII 2D EXAMPLES OF GBSK

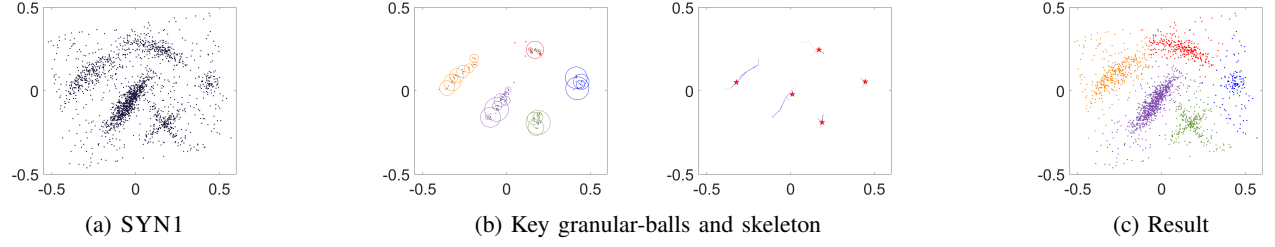


Fig. 12: A 2D example: (a) Original data of SYN1. (b) The left shows the key balls, and the right shows the skeleton of SYN1. (c) Clustering result of GBSK.

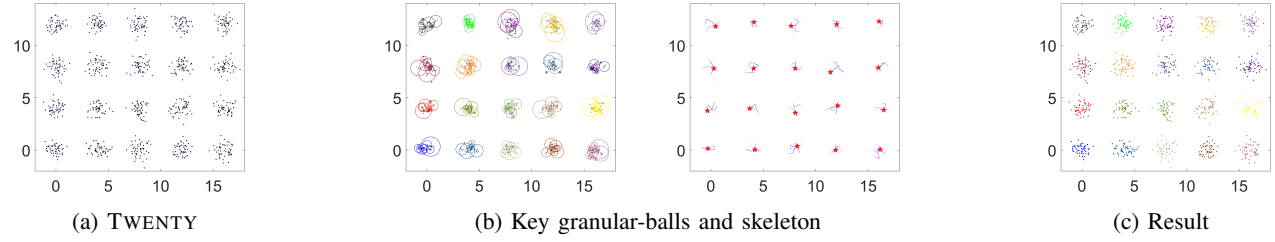


Fig. 13: 2D example on TWENTY.

APPENDIX VIII
PARAMETER STUDY RESULTS

TABLE XII: Impact of s under $[*, \frac{1}{\sqrt{n}}, 10k, k]$ setting for GBSK on AGC100M, averaged over 50 runs with 95% confidence intervals.

s	GBSK runtime (s)	Step 1 load (s)	Step 2 (s)	Step 3 (s)	Step 4 (s)	Step 5 load (s)	ARBC	W	ACC	ARI	AMI
5	387.2 ± 13.6	1.4 ± 0.0	3.3 ± 0.3	0.03 ± 0.00	0.001 ± 0.000	382.5 ± 13.5	50.5 ± 3.7	20.4 ± 0.3	0.953 ± 0.002	0.948 ± 0.003	0.986 ± 0.002
10	375.8 ± 9.7	4.6 ± 3.7	5.3 ± 0.4	0.04 ± 0.00	0.001 ± 0.001	365.8 ± 7.6	84.7 ± 6.4	23.3 ± 0.8	0.951 ± 0.001	0.947 ± 0.002	0.987 ± 0.000
15	447.1 ± 4.8	4.1 ± 0.0	8.0 ± 0.6	0.05 ± 0.00	0.001 ± 0.000	434.9 ± 4.6	127.1 ± 8.6	25.5 ± 0.8	0.952 ± 0.001	0.949 ± 0.002	0.987 ± 0.000
20	454.5 ± 6.9	5.6 ± 0.1	11.9 ± 0.7	0.05 ± 0.00	0.001 ± 0.001	436.9 ± 6.8	181.0 ± 9.7	26.5 ± 0.9	0.951 ± 0.001	0.946 ± 0.002	0.986 ± 0.000
25	466.5 ± 3.8	6.8 ± 0.0	13.9 ± 0.7	0.05 ± 0.00	0.001 ± 0.000	445.7 ± 3.8	217.9 ± 9.8	28.4 ± 1.0	0.951 ± 0.001	0.946 ± 0.002	0.986 ± 0.000
30	491.2 ± 7.3	8.5 ± 0.1	16.7 ± 0.5	0.06 ± 0.00	0.001 ± 0.001	465.9 ± 7.2	253.6 ± 7.9	29.9 ± 0.9	0.951 ± 0.001	0.946 ± 0.001	0.986 ± 0.000
35	412.2 ± 11.9	9.6 ± 0.1	22.4 ± 1.1	0.07 ± 0.00	0.001 ± 0.000	380.2 ± 12.3	297.5 ± 12.3	31.2 ± 1.4	0.947 ± 0.008	0.943 ± 0.008	0.985 ± 0.002
40	503.0 ± 6.3	11.2 ± 0.1	23.4 ± 1.1	0.06 ± 0.00	0.001 ± 0.000	468.2 ± 6.1	352.5 ± 14.7	32.7 ± 1.5	0.951 ± 0.001	0.946 ± 0.002	0.986 ± 0.000
45	421.3 ± 5.2	12.3 ± 0.0	31.3 ± 1.2	0.08 ± 0.00	0.001 ± 0.000	377.6 ± 4.8	397.4 ± 13.5	34.2 ± 1.5	0.952 ± 0.001	0.946 ± 0.002	0.986 ± 0.000
50	524.6 ± 6.0	13.7 ± 0.0	29.0 ± 1.1	0.07 ± 0.00	0.001 ± 0.000	481.8 ± 6.0	435.2 ± 15.8	34.4 ± 1.9	0.951 ± 0.001	0.945 ± 0.002	0.986 ± 0.000
55	443.5 ± 7.0	15.1 ± 0.0	39.0 ± 1.5	0.10 ± 0.00	0.001 ± 0.000	389.2 ± 6.7	480.0 ± 16.9	36.4 ± 2.1	0.951 ± 0.001	0.946 ± 0.002	0.986 ± 0.000
60	545.1 ± 6.1	16.5 ± 0.0	34.9 ± 1.2	0.08 ± 0.00	0.001 ± 0.001	493.5 ± 6.2	513.2 ± 17.1	35.9 ± 1.8	0.947 ± 0.006	0.942 ± 0.006	0.985 ± 0.002
65	467.5 ± 12.7	18.1 ± 0.2	46.6 ± 2.3	0.11 ± 0.01	0.001 ± 0.000	402.5 ± 11.2	562.2 ± 19.4	37.4 ± 2.0	0.951 ± 0.001	0.945 ± 0.002	0.986 ± 0.000
70	568.3 ± 9.3	19.4 ± 0.0	41.7 ± 1.3	0.09 ± 0.00	0.001 ± 0.000	507.0 ± 9.1	616.3 ± 18.7	39.2 ± 2.4	0.951 ± 0.001	0.944 ± 0.002	0.986 ± 0.000
75	530.1 ± 12.9	21.4 ± 0.2	45.9 ± 1.8	0.09 ± 0.01	0.001 ± 0.000	462.5 ± 13.0	661.3 ± 19.6	38.8 ± 2.7	0.951 ± 0.001	0.945 ± 0.002	0.986 ± 0.000
80	586.0 ± 9.2	22.2 ± 0.1	47.6 ± 1.5	0.09 ± 0.00	0.001 ± 0.000	516.0 ± 9.1	697.3 ± 20.9	38.9 ± 2.0	0.948 ± 0.005	0.942 ± 0.005	0.985 ± 0.002
90	589.3 ± 11.3	24.7 ± 0.0	52.6 ± 1.4	0.10 ± 0.01	0.001 ± 0.000	511.7 ± 11.3	782.9 ± 20.1	42.1 ± 3.5	0.948 ± 0.004	0.941 ± 0.003	0.985 ± 0.001
100	616.7 ± 12.2	27.9 ± 0.2	60.3 ± 1.6	0.10 ± 0.00	0.001 ± 0.000	528.2 ± 11.5	882.1 ± 22.5	41.5 ± 2.6	0.951 ± 0.001	0.945 ± 0.002	0.986 ± 0.000

“Step 1 load (s)” and “Step 5 load (s)” refer to the time required for loading batches from the physical data file and the processing step time, respectively. |ARBC|: number of centers in $ARBC$. “W”: number of key balls. $k = 17$. $\frac{1}{\sqrt{n}} = 1 \times 10^{-4}$. All values in the table are reported as: mean $\pm$ 95% confidence interval. Bold values mark the default parameter setting of AGBSK.

TABLE XIII: Impact of α under $[30, *, 10k, k]$ setting for GBSK on AGC100M, averaged over 50 runs with 95% confidence intervals.

α	GBSK runtime (s)	Step 1 load (s)	Step 2 (s)	Step 3 (s)	Step 4 (s)	Step 5 load (s)	ARBC	W	ACC	ARI	AMI
1×10^{-6}	389.5 ± 17.9	3.4 ± 1.2	1.0 ± 0.0	0.06 ± 0.00	0.001 ± 0.001	384.9 ± 16.8	398.3 ± 10.6	34.6 ± 1.8	0.943 ± 0.007	0.935 ± 0.011	0.980 ± 0.004
1×10^{-5}	435.9 ± 6.3	3.5 ± 0.0	3.8 ± 0.1	0.07 ± 0.00	0.001 ± 0.000	428.5 ± 6.3	335.6 ± 11.8	35.5 ± 1.8	0.952 ± 0.001	0.947 ± 0.002	0.987 ± 0.000
1×10^{-4}	491.2 ± 7.3	8.5 ± 0.1	16.7 ± 0.5	0.06 ± 0.00	0.001 ± 0.001	465.9 ± 7.2	253.6 ± 7.9	29.9 ± 0.9	0.951 ± 0.001	0.946 ± 0.001	0.986 ± 0.000
1×10^{-3}	636.3 ± 13.6	53.7 ± 0.2	123.0 ± 7.7	0.07 ± 0.00	0.001 ± 0.000	459.4 ± 9.9	200.5 ± 11.8	38.5 ± 2.1	0.950 ± 0.002	0.942 ± 0.003	0.986 ± 0.001
1×10^{-2}	2343.6 ± 102.0	515.1 ± 2.0	1326.4 ± 97.4	0.07 ± 0.00	0.001 ± 0.000	501.9 ± 14.6	171.8 ± 9.8	44.3 ± 2.9	0.949 ± 0.004	0.943 ± 0.007	0.986 ± 0.001

TABLE XIV: Impact of M under $[30, \frac{1}{\sqrt{n}}, *, k]$ setting for GBSK on MNIST8M, averaged over 50 runs with 95% confidence intervals.

M	GBSK runtime (s)	Step 1 load (s)	Step 2 (s)	Step 3 (s)	Step 4 (s)	Step 5 load (s)	ARBC	W	ACC	ARI	AMI
1k	187.0 ± 3.6	4.0 ± 0.1	18.7 ± 0.2	0.19 ± 0.00	0.002 ± 0.001	164.0 ± 3.6	299.8 ± 0.4	90.0 ± 2.1	0.409 ± 0.006	0.221 ± 0.005	0.352 ± 0.005
2k	194.1 ± 4.7	4.0 ± 0.0	21.4 ± 0.2	0.20 ± 0.01	0.002 ± 0.000	168.3 ± 4.7	300.0 ± 0.0	91.2 ± 2.7	0.402 ± 0.008	0.230 ± 0.007	0.376 ± 0.006
3k	188.7 ± 3.6	3.9 ± 0.1	21.7 ± 0.1	0.19 ± 0.01	0.002 ± 0.000	162.8 ± 3.6	300.0 ± 0.0	88.5 ± 2.1	0.406 ± 0.010	0.240 ± 0.008	0.381 ± 0.007
4k	192.7 ± 3.9	4.0 ± 0.0	23.5 ± 0.2	0.19 ± 0.00	0.002 ± 0.000	164.9 ± 3.8	300.0 ± 0.0	89.4 ± 2.2	0.397 ± 0.010	0.228 ± 0.009	0.374 ± 0.008
5k	212.4 ± 4.3	4.4 ± 0.1	26.0 ± 0.4	0.21 ± 0.00	0.002 ± 0.000	181.7 ± 4.1	299.7 ± 0.4	90.2 ± 1.9	0.397 ± 0.011	0.234 ± 0.010	0.383 ± 0.008
6k	212.9 ± 5.5	4.4 ± 0.1	27.1 ± 0.4	0.21 ± 0.01	0.002 ± 0.000	181.1 ± 5.5	299.7 ± 0.4	89.5 ± 2.7	0.405 ± 0.010	0.239 ± 0.009	0.389 ± 0.007
7k	215.6 ± 5.4	4.4 ± 0.1	29.8 ± 0.4	0.21 ± 0.01	0.003 ± 0.001	181.0 ± 5.4	299.8 ± 0.3	90.1 ± 2.5	0.390 ± 0.012	0.228 ± 0.012	0.380 ± 0.009
8k	207.9 ± 5.1	4.1 ± 0.0	28.8 ± 0.3	0.21 ± 0.01	0.002 ± 0.000	174.6 ± 5.1	299.7 ± 0.5	91.5 ± 2.8	0.388 ± 0.012	0.229 ± 0.011	0.380 ± 0.009
9k	209.8 ± 3.4	4.1 ± 0.0	29.0 ± 0.3	0.21 ± 0.00	0.002 ± 0.000	176.4 ± 3.3	300.0 ± 0.0	93.2 ± 1.8	0.387 ± 0.012	0.223 ± 0.012	0.379 ± 0.009
10k	209.3 ± 3.8	4.1 ± 0.0	29.2 ± 0.3	0.21 ± 0.00	0.002 ± 0.000	175.7 ± 3.8	299.8 ± 0.3	92.6 ± 2.0	0.389 ± 0.012	0.231 ± 0.010	0.383 ± 0.009
11k	208.6 ± 4.9	4.1 ± 0.0	29.8 ± 0.3	0.21 ± 0.01	0.002 ± 0.000	174.3 ± 5.0	300.0 ± 0.0	91.5 ± 2.8	0.392 ± 0.013	0.226 ± 0.012	0.382 ± 0.010
12k	208.7 ± 4.3	4.1 ± 0.0	31.7 ± 0.3	0.20 ± 0.00	0.002 ± 0.000	172.5 ± 4.4	300.0 ± 0.0	90.7 ± 2.2	0.380 ± 0.011	0.219 ± 0.010	0.375 ± 0.008
13k	215.9 ± 4.3	4.1 ± 0.0	35.7 ± 0.3	0.21 ± 0.00	0.002 ± 0.000	175.7 ± 4.3	299.8 ± 0.3	93.0 ± 2.3	0.393 ± 0.012	0.231 ± 0.010	0.382 ± 0.008
14k	209.2 ± 5.2	4.1 ± 0.0	34.4 ± 0.3	0.20 ± 0.01	0.002 ± 0.000	170.4 ± 5.2	299.7 ± 0.4	90.6 ± 2.6	0.391 ± 0.013	0.229 ± 0.010	0.383 ± 0.009
15k	204.0 ± 4.4	4.0 ± 0.0	33.7 ± 0.2	0.20 ± 0.00	0.002 ± 0.000	165.9 ± 4.4	300.0 ± 0.0	90.8 ± 2.6	0.394 ± 0.010	0.230 ± 0.009	0.382 ± 0.008
20k	204.7 ± 5.1	4.0 ± 0.0	35.2 ± 0.2	0.19 ± 0.01	0.002 ± 0.000	165.1 ± 5.0	299.8 ± 0.3	90.2 ± 2.9	0.391 ± 0.011	0.235 ± 0.010	0.384 ± 0.009
25k	213.9 ± 4.7	4.0 ± 0.0	44.4 ± 0.5	0.20 ± 0.00	0.002 ± 0.000	165.1 ± 4.6	300.0 ± 0.0	89.6 ± 2.5	0.371 ± 0.011	0.207 ± 0.012	0.369 ± 0.009
30k	213.8 ± 3.7	3.9 ± 0.0	45.2 ± 0.2	0.20 ± 0.00	0.002 ± 0.000	164.3 ± 3.7	299.8 ± 0.4	90.1 ± 2.2	0.397 ± 0.015	0.229 ± 0.014	0.382 ± 0.011
35k	218.0 ± 3.9	4.0 ± 0.0	48.2 ± 0.3	0.19 ± 0.00	0.002 ± 0.000	165.5 ± 3.9	300.0 ± 0.0	90.4 ± 2.3	0.392 ± 0.010	0.216 ± 0.010	0.372 ± 0.009
40k	223.4 ± 5.8	4.0 ± 0.0	58.5 ± 0.2	0.19 ± 0.01	0.002 ± 0.000	160.6 ± 5.9	299.8 ± 0.3	87.6 ± 3.4	0.387 ± 0.012	0.220 ± 0.010	0.372 ± 0.008
50k	227.8 ± 5.2	4.0 ± 0.0	63.8 ± 0.3	0.19 ± 0.01	0.002 ± 0.000	159.7 ± 5.2	300.0 ± 0.0	87.3 ± 3.0	0.393 ± 0.011	0.222 ± 0.012	0.375 ± 0.009
70k	247.5 ± 5.3	4.0 ± 0.1	85.1 ± 0.6	0.19 ± 0.01	0.002 ± 0.000	158.1 ± 5.3	299.5 ± 0.6	85.7 ± 3.3	0.403 ± 0.012	0.228 ± 0.011	0.383 ± 0.008

APPENDIX IX DETAILED PARAMETER SENSITIVITY ANALYSIS

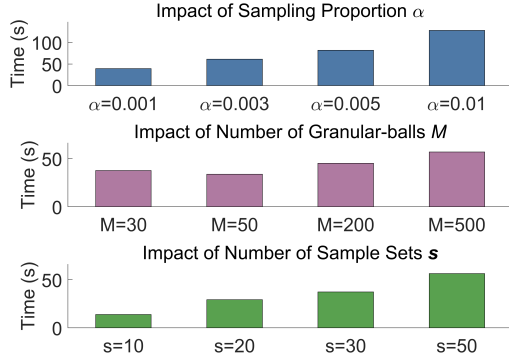


Fig. 14: Runtime under varying parameter settings for GBSK on the N-BAIoT dataset.

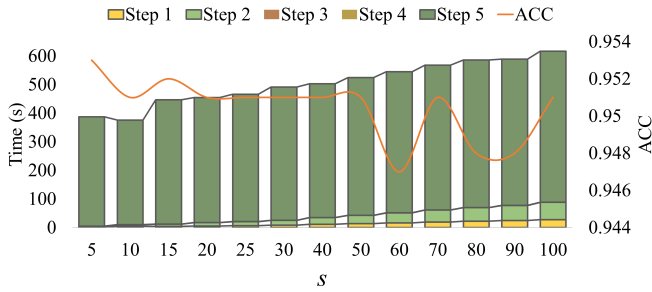


Fig. 15: **Impact of s** on the time (in seconds) and accuracy for each step under $[*, \frac{1}{\sqrt{n}}, 10k, k]$ setting for GBSK on AGC100M, averaged over 50 runs.

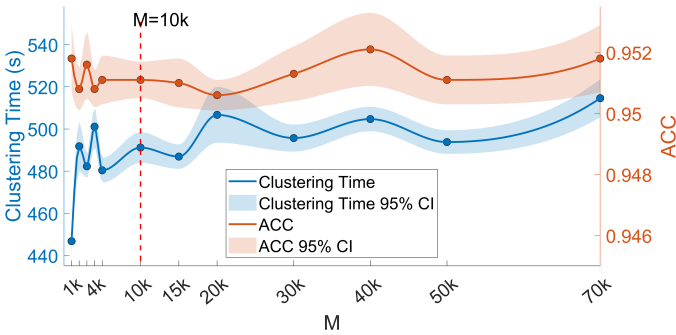


Fig. 16: **Impact of M** on runtime (s) and accuracy under $[30, \frac{1}{\sqrt{n}}, *, k]$ setting for GBSK on AGC100M, averaged over 50 runs with 95% confidence intervals. The vertical dashed red line highlights $M = 10k = 170$.

This appendix provides a granular analysis of the impact of the three governing hyperparameters, sampling ratio (α), resampling iterations (s), and granular-ball count (M), on the computational efficiency and clustering stability of the proposed GBSK method.

A. Impact of Sampling Ratio (α)

The sampling ratio α dictates the fidelity of the local approximation relative to the global dataset. As α increases,

the cardinality of each subsample grows linearly, which in turn imposes a polynomial increase on the granular-ball generation process (Step 2). Empirical evidence from Table XIII confirms this theoretical relationship: increasing α from 10^{-4} to 10^{-2} results in a precipitous rise in the execution time of the sampling and ball-generation stages. Consequently, for very-large-scale datasets ($n > 10^7$), the proposed adaptive heuristic $\alpha = n^{-1/2}$ is critical for maintaining sub-linear scalability without compromising the structural integrity of the extracted skeleton. More generally, effective clustering requires that the expected subsample size significantly exceed the target cluster number, i.e., $\alpha n \gg k$ so that each cluster is adequately represented in the sampled subsets. Within this regime, GBSK exhibits stable behavior without requiring precise tuning of α .

B. Impact of Resampling Iterations (s)

The parameter s controls the ensemble size, directly influencing the robustness of the skeleton against sampling bias. As illustrated in Fig. 15 and Table XII, increasing s enhances the coverage of the density landscape, reducing the variance of the mode estimates. However, this comes at a linear computational cost. Our experiments reveal a clear saturation point at approximately $s = 30$. Beyond this threshold, the marginal gain in clustering accuracy diminishes, while the runtime continues to climb. Conversely, setting $s < 10$ increases the risk of skeleton fragmentation, where insufficient Key Balls are generated to form a connected topology. Thus, $s = 30$ represents the optimal trade-off between statistical stability and computational efficiency.

C. Impact of Granular Resolution (M)

The parameter M determines the resolution of the intermediate granular-ball representation. Experimental results on MNIST8M (Table XIV) demonstrate a wide stability plateau: GBSK maintains high clustering accuracy for M ranging from $5k$ to $70k$. Crucially, increasing M beyond $10k$ yields negligible improvements in accuracy while significantly inflating the memory footprint and the complexity of the granular-ball construction step ($O(M^2)$). These observations empirically validate the mild scaling constraint $k < M \ll n$, under which GBSK operates in a stable regime. In practice, setting $M = 10k$ places the method near the center of the stability plateau, achieving a favorable trade-off between structural resolution and computational efficiency without unnecessary over-refinement.

Collectively, these sensitivity analyses substantiate the design of the proposed AGBSK variant. The derived default configuration, i.e., $\alpha = n^{-1/2}$, $s = 30$, and $M = 10k$, lies within the sweet spot of the hyperparameter space for a diverse array of datasets, effectively eliminating the need for manual tuning in practical applications.

APPENDIX X
ALGORITHMS IN ABLATION STUDY

Algorithm 6 GBSK without sampling

Input: dataset P ($n \times d$ matrix), category number k

Output: clustering labels of points in P

```

1: /* — Step 1. Granular-ball Generation          — */
2:  $balls$  = obtain  $10k$  granular-balls by Algorithm 1 or 2
3: /* — Step 2. Skeleton Construction            — */
4:  $[rootIDs, parentIDs]$  = construct a forest for  $balls$  by
   Algorithm 4
5: for  $i = 1$  to  $|rootIDs|$  do
6:    $j = rootIDs(i)$ 
7:   assign a new label to the root granular-ball  $balls_j$ 
8: end for
9: for  $\forall balls_j \in balls$  do
10:  assign a label to  $balls_j$  by (11)
11: end for
12: /* — Step 3. Final Cluster Assignment          — */
13: for each  $p \in P$  do
14:  // assign labels to the remaining points
15:  assign a label to  $p$  by (12)
16: end for

```

Algorithm 7 GBSK without representative balls

Input: dataset P ($n \times d$ matrix), category number k

Output: clustering labels of points in P

```

1: /* — Step 1. Random Sampling                  — */
2:  $v = \sqrt{n}$  // sample set size
3: obtain sample sets  $P^{(1)}, P^{(2)}, \dots, P^{(30)}$  from  $P$ 
4: /* — Step 2. Granular-ball Generation          — */
5:  $aggBalls \leftarrow \emptyset$ 
6: for each  $P^{(i)}$  do
7:   $balls^{(i)} \leftarrow$  obtain  $10k$  granular-balls by Algorithm 1 or
    2
8:  push  $balls^{(i)}$  into  $aggBalls$ 
9: end for
10: /* — Step 3. Skeleton Construction            — */
11:  $[rootIDs, parentIDs]$  = construct a forest for  $aggBalls$ 
    by Algorithm 4
12: for  $i = 1$  to  $|rootIDs|$  do
13:   $j = rootIDs(i)$ 
14:  assign a new label to the root ball  $aggBalls_j$ 
15: end for
16: for  $\forall aggBalls_j \in aggBalls$  do
17:  assign a label to  $aggBalls_j$  by (11)
18: end for
19: /* — Step 4. Final Cluster Assignment          — */
20: for each  $p \in P$  do
21:  // assign labels to the remaining points
22:  assign a label to  $p$  by (12)
23: end for

```

Algorithm 8 GBSK without granular-balls

Input: dataset P ($n \times d$ matrix), category number k

Output: clustering labels of points in P

```

1: /* — Step 1. Random Sampling                  — */
2:  $v = \sqrt{n}$  // sample set size
3: obtain sample sets  $P^{(1)}, P^{(2)}, \dots, P^{(30)}$  from  $P$ 
4: /* — Step 2. Identifying Local Peaks          — */
5:  $aggPeaks \leftarrow \emptyset$ 
6: for each  $P^{(i)}$  do
7:  //select top  $k$  peaks similar to Algorithm 3
8:   $localPeaks^{(i)} =$  top  $k$  peaks by DPeak [6]
9:  push  $localPeaks^{(i)}$  into  $aggPeaks$ 
10: end for
11: /* — Step 3. Skeleton Construction            — */
12: cluster  $aggPeaks$  by DPeak into  $k$  clusters
13: /* — Step 4. Final Cluster Assignment          — */
14: for each  $p \in P$  do
15:  // assign labels to the remaining points
16:   $q =$  find  $p$ 's nearest neighbor in  $aggPeaks$ 
17:  assign the label of  $q$  to  $p$ 
18: end for

```

APPENDIX XI

DESCRIPTION OF MAIN SYMBOLS

Symbol	Description
P	Original dataset.
n	Number of instances of P .
$P^{(i)}$	i -th sample set.
$p_j^{(i)}$	j -th point in $P^{(i)}$
v	Number of points in a sample set.
$f(x)$	density function of distribution $\mathcal{F}$
$\hat{f}_X(x)$	kernel density estimator of f on samples X
$B_j^{(i)}$	j -th granular-ball in $P^{(i)}$.
B	A granular-ball.
$x_{i,j}$	j -th point in B_i .
$ X_B $	Number of points within B .
$B.E$	Points inside B .
$B.c$	Center of B .
$B.\rho$	Density of B .
$B.r$	Radius of B .
$B.\phi$	Distribution measure of B .
ϵ	Smoothing constant in ρ_i .
τ	Regularization constant in ϕ_i .
$\Phi(B)$	Condition to split B .
$meanR$	Mean value of the radii of all balls.
$medianR$	Median radius of all balls.
$repBalls^{(i)}$	Representative balls identified from $P^{(i)}$.
$repBallCenters^{(i)}$	Center set of $repBalls^{(i)}$.
$ARBC$	Aggregated centers of all representative balls.
$keyBalls$	Key balls identified from $ARBC$.
$keyBall_i$	i -th ball of $keyBalls$.
W	Number of key balls.
$\delta_j^{(i)}$	Distance to $ball_j^{(i)}$'s nearest ball of higher density.
$\gamma_j^{(i)}$	Value for selecting peaks.
$Root(keyBall_i)$	Root ball of $keyBall_i$ in the skeleton.
$rootBalls$	Center balls of clusters selected from key balls.
$label(ball)$	Category label of $ball$.
$label(p)$	Category label of point p .
s	Sampling count of GBSK.
α	Sample proportion of GBSK.
M	Rough number of granular-balls.
k	Number of clusters.
$localPeaks^{(i)}$	Peak points selected from $P^{(i)}$.
$aggPeaks$	Aggregated peaks from all local peaks.

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